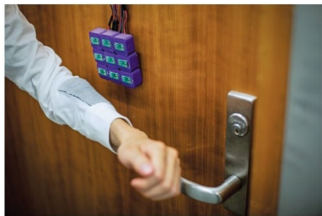


fabric improved the performance of the printed devices. The versatility of this process allowed the researchers to design not only single transistors but all-printed integrated electronic circuits combining active and passive components.

"The printed components are flexible, washable and require low power—essential requirements for applications in wearable electronics," said PhD student Tian Carey, the paper's first author.

## Store data in your clothes without electronics

Computer scientists have created fabrics and fashion accessories that can store data from security codes to identification tags without requiring any on-board electronics or sensors. They leveraged previously unexplored magnetic properties of off-the-shelf conductive thread. The data can be read using an instrument embedded in existing smartphones to enable navigation apps.



Using magnetic properties of conductive thread, University of Washington researchers are able to store data in fabric. In this example, the code to unlock a door is stored in a fabric patch and read by an array of magnetometers (Credit: Dennis Wise/University of Washington)

"This is a completely electronic-free design, which means you can iron the smart fabric or put it in the washer and dryer," said senior author Shyam Gollakota, associate professor in the Paul G. Allen School of Computer Science & Engineering. "You can think of the fabric as a hard disk—you're actually doing this data storage on the clothes you're wearing."

Researchers realised that magnetic properties of the conductive thread can be manipulated to store either digital data or visual information like letters or numbers. This data can be read by a magnetometer—an inexpensive instrument that measures the direction and strength of magnetic fields and is embedded in most smartphones.

"We are using something that already exists on a smartphone and uses almost no power, so the cost of reading this type of data is negligible," said Gollakota. In one example, they stored the passcode to an electronic door

lock on a patch of conductive fabric sewn to a shirt cuff. They unlocked the door by waving the cuff in front of an array of magnetometers.

## Smart ring to control smartwatch precisely

With mobile devices such as Apple Watch, interactive screens are so small that these are difficult to use. Computer scientists at Saarland University have now developed an alternative called 'DeformWear'. It's a tiny switch built into a ring for example, and worn on the body. It can be moved in all directions, pressed, pinched, and pushed toward the right, left, up and down.



Researchers' prototype can be controlled precisely and discreetly using minimal movements (Credit: Saarland University)

The sensor developed by researchers has a diameter of only 10 millimetres—about the size of a pea—and can be deformed like a balloon. From the inside, an infrared light-emitting diode illuminates the deformable membrane. The light is reflected, and measured by four photodiodes. These measurements can then be used to calculate how the sensor is being deformed.

To test their idea, researchers integrated this sensor into a ring, a bracelet and a charm, which was hardly larger than a 50-cent piece. The challenge was to develop gestures and use them to control mobile devices. The researchers did this for a smartwatch and for virtual reality glasses. They also defined motion sequences to control a television and to play music without having to look at the device.

## Self-learning chip

Intel has developed a first-of-its-kind self-learning neuromorphic chip, codenamed Loihi, that mimics how the brain functions by learning to operate based on various modes of feedback from the environment. This extremely energy-efficient chip, which uses



Self-learning neuromorphic chip